

CSE340: Computer Architecture

Assignment 4

Chapter 4

Question 01

Design a 16 X 132 register file.

Note: You must label all the input and output pins

Question 02

Why does the ALU control unit take the following bits as input?

Instruction[30, 14-12]

Question 03

What is the significance of the Zero pin in the ALU?

Question 04

Suppose you have a single instruction;

Add X21, X22, X23

Register read = 20s

Instruction Fetch = 20s

Memory Operation = 40s

Register write = 20s

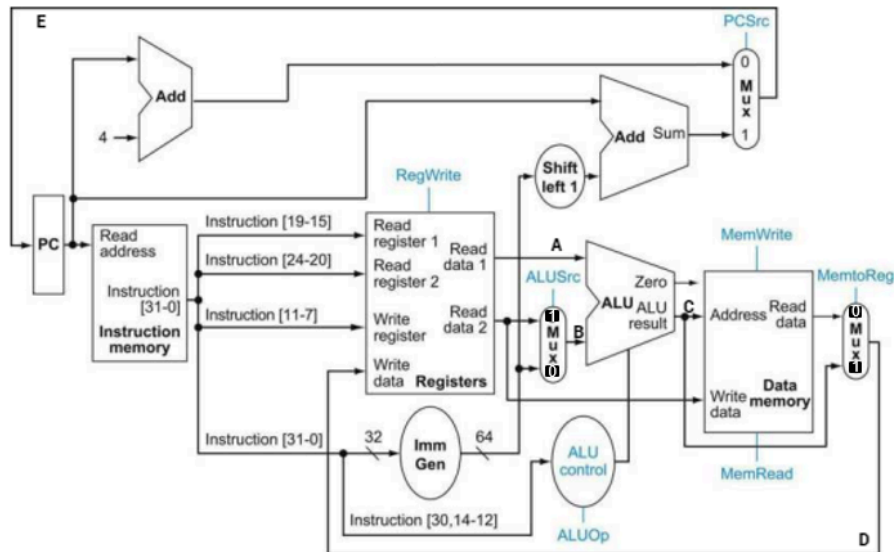
ALU Operation = 40s

Suppose you have a pipelined setup and a single-cycle setup,
Which one will run the instruction faster?

Question 05

Modify the BEQ datapath so that it works for BNE.

Question 06



Write the control signals for the following instruction:

Ld X5, 10(X21);

- i. ALU Src;
- ii. RegWrite;
- iii. MemRead;
- iv. MemWrite;
- v. MemtoReg.

Question 07

Explain control hazard in your own words.

Question 08

Why do we keep instruction and data memory separate?